Vikas Ekka

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Summary

Software Engineer with **3+ years** of experience building scalable, high-performance applications in a fast-paced product environment. Proficient in developing robust backend systems with **Node.js, Express.js** and managing large-scale data operations using **BigQuery**, complemented by modern front-end development in **Angular**.

Skills

NodeJS, Angular, BigQuery, ExpressJs, Git, RabbitMQ, RxJs, Firebase, Sequelize, Redis, Google Cloud Storage Buckets, REST APIs, TypeScript, Restful API's, DBMS, HTML

Professional Experience

Software Developer — *Indore, MP*
MSG91, WALKOVER WEB SOLUTIONS PVT. LTD. — *Jun 2022 – Present*

- Engineered a scalable reports **microservice** executing **ETL** on **10M+ records** daily to drive a **real-time analytics** dashboard; leveraged **Looker Studio** for direct visualization (reducing UI effort by 30%) and implemented automated email/Slack alerts for ETL failures and data anomalies to maintain system reliability.
- Built a queue-backed webhook delivery system processing 100K+ daily events with integrations to **WebEngage**, **MoEngage**, and **CleverTap**; implemented **exponential retry mechanism**, **rate limiting**, and **auto-pause safeguards**, achieving high delivery reliability while protecting client endpoints from overload.
- Developed a customer churn prediction feature using **Google ML** and **BigQuery**, which improved customer loyalty by 15% over six months by providing actionable insights into customer behavior.
- Designed and implemented an anomaly detection system using **Node.js** and **machine learning models**, training supervised models on large-scale **BigQuery** datasets for real-time traffic analysis, reducing bot attacks by 40% and onboarding of key enterprise clients through enhanced security capabilities.
- Created **reusable Angular components** integrated with **NgRx** for efficient **state management**, significantly enhancing code maintainability and boosting development speed by 40%.
- Built a real-time billing analytics system with **Node.js**, **Express.js**, and **RESTful APIs**, processing **10M+ records** monthly for **500+ clients**; automated invoice computation and vendor reconciliation, cutting manual effort by ~60% and achieving ~99% billing accuracy.
- Migrated RabbitMQ consumers to **Kubernetes** with horizontal pod auto-scaling and health checks, improving system availability to 99.9% and enabling seamless scaling under peak traffic.
- Implemented **Redis caching** with TTL for high-frequency polling endpoints, preventing repeated identical DB queries from client reload abuse, reducing DB load by 40% and improving API response time by 35%.
- Resolved a production memory spike in a **RabbitMQ**-based **Node.js** microservice where oversized payloads (500K+ records per message) raised heap usage from ~350MB to 1.5GB, causing OOM crashes. Re-architected messaging to use 1K-record batches and applied `channel.prefetch(5)` for backpressure control, reducing peak memory by ~65%, eliminating crashes, and supporting 2× higher traffic.
- Developed a **File System** and sharing utility using **Google Buckets** and **Firebase**, enabling users to upload files and generate shareable links, handling over **10,000 file** transactions monthly with 99.9% uptime.
- Optimized **CI/CD pipelines** and enforced structured **peer code reviews** within an Agile workflow, reducing release cycle time, improving sprint velocity by 25%, and decreasing production defects by 30%.

Education

Bhilai Institute of Technology — *Aug 2016 – Aug 2020*
B.Tech (Mechanical Engineering)

Key Strengths

strong analytical thinker, Team Work, Leadership, Communication